INFF-Net: Rotation-Aware Remote Sensing Object Detection Based on Implicit Neural Feature Field

Yan Dong, Xiangyun Li, Guangshuai Gao, *Member, IEEE*, Jun Zhang, Jijie Jia, Yunqi Shang, and Chunlei Li

***Abstract*—Remote sensing objects are characterized by arbitrary orientations, complex backgrounds, and high aspect ratios (ARs). Most existing methods rely on explicit discrete convolutions to construct feature networks, which suffer from limitations such as insufficient rotation-invariant representation, background noise introduced by uniform sampling of regions of interest (RoIs), and difficulty in handling objects with extreme ARs. To address these issues, this article proposes implicit neural feature fields (INFF-Net), a rotation-aware detection network based on INFF-Net. It comprises three core components: the implicit rotation feature field (IRFF) constructs a spatial-angular continuous representation and introduces a rotation consistency regularization loss to impose geometric constraints, thereby mitigating the pose sensitivity and angular boundary discontinuity associated with discrete convolution. The importance-guided sparse sampling modulation (ISSM) module adaptively modulates RoI features through sparse coarse sampling and importance-score-based refined sampling, effectively suppressing background noise. The adaptive shape-aware feature enhancement (ASFE) module dynamically adjusts the number of sampling points based on objects' ARs and employs elliptical sampling to conform to the contours, improving the detection accuracy of elongated objects. The method achieves the mean average precision (mAP) scores of 77.87%, 66.67%, and 97.90% on the DOTA-v1.0, DIOR-R, and HRSC2016 datasets, respectively, obtaining significant improvements of 3.92%, 4.48%, and 1.40% over the baseline, demonstrating the effectiveness and generalization capability.**

***Index Terms*—Adaptive sampling, implicit neural feature field (INFF-Net), remote sensing images, rotated object detection, rotation-aware encoding.**

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Yan Dong, Xiangyun Li, Guangshuai Gao, Jijie Jia, Yunqi Shang, and Chunlei Li are with the School of Information and Communication Engineering, Zhongyuan University of Technology, Zhengzhou 450007, China (e-mail: dy@zut.edu.cn; 2024019010@zut.edu.cn; 6911@zut.edu.cn; 2024019009@zut.edu.cn; shangyunqi114@163.com; lichunlei1979@zut.edu.cn).

Jun Zhang is with Wuzhi Second Yellow River Affairs Bureau, Jiaozuo Yellow River Affairs Bureau, Jiaozuo 454950, China (e-mail: 281700488@qq.com).

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I. Introduction

REMOTE sensing object detection aims to automatically perform precise localization and classification of objects from aerial and satellite imagery [1], playing a critical supporting role in fields such as land monitoring, military reconnaissance, and disaster relief [2], [3]. Unlike objects in natural scenes, objects in remote sensing imagery typically exhibit characteristics such as arbitrary orientation, high aspect ratios (ARs), and complex backgrounds [4], [5]. Traditional detection methods with horizontal bounding boxes struggle to effectively align with the precise contours of rotated objects. Especially in dense scenes, these methods are prone to present overlaps, boundary shifts, and a high number of false positives, making it difficult to meet the practical demands of high-precision remote sensing interpretation [6], [7]. In contrast, rotated object detection methods can precisely match the rotated pose and geometric shape of objects, and have become an important research paradigm [8].

Driven by deep learning technologies, remote sensing rotated object detection has undergone unprecedented methodological evolution. Based on different network architectures, current mainstream methods can be categorized into three types: two-stage methods, represented by Faster R-CNN [9] and Oriented R-CNN [10], which generate candidate regions via a region proposal network followed by fine-grained regression and classification [11]. Single-stage methods such as YOLO [12] and RetinaNet [13] directly regress rotated bounding boxes on feature maps to achieve faster inference speeds. Transformer-based end-to-end methods utilize an ensemble prediction mechanism to produce end-to-end outputs for rotated objects, thereby bypassing anchor design and postprocessing workflows [14].

Despite the progress achieved by the aforementioned methods, they generally rely on explicit discrete convolutions for feature extraction and still suffer from three inherent limitations when confronting challenges such as arbitrary orientations, extreme ARs, and complex backgrounds in remote sensing scenarios: 1) insufficient rotation-aware feature representation, where fixed-geometry convolution kernels fail to adapt to arbitrary object orientations and conventional angle encoding suffers from boundary discontinuity, leading to inconsistent feature representations across different poses; 2) poor foreground discriminability in complex backgrounds, where uniform sampling with fixed density struggles to distinguish foreground objects from background clutter, inevitably

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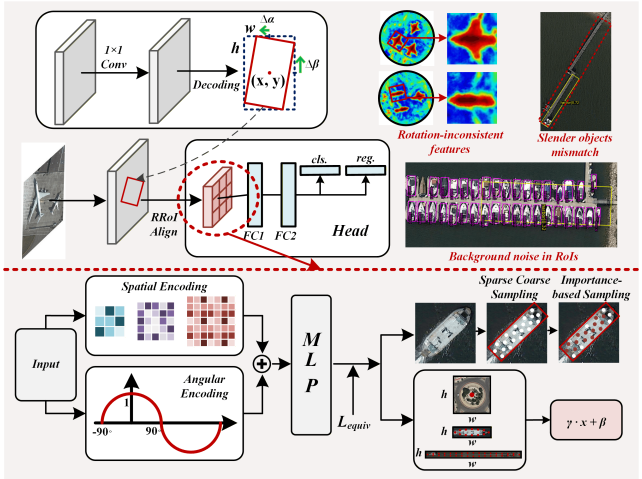


Fig. 1. (Top) Comparison of baseline and (bottom) proposed architectures. Three bottlenecks of the baseline are illustrated: rotation-inconsistent features from fixed convolution kernels, background noise from uniform RoI sampling, and shape mismatch of square receptive fields for slender objects. Our method solves them with IRFF, ISSM, and ASFE modules.

introducing substantial noise; and 3) limited adaptive modeling capability for high-aspect-ratio objects, where fixed sampling points and square receptive fields cannot accommodate the geometric variations of elongated objects, resulting in inadequate contour feature extraction. As illustrated in the upper half of Fig. 1, these limitations manifest in the baseline architecture as follows. Fixed convolution kernels produce inconsistent feature responses across rotation angles, fixed uniform sampling fails to discriminate foreground from complex backgrounds, and square receptive fields struggle to cover the spatial extent of slender objects.

To address these issues, this article proposes a rotation-aware remote sensing object detection method based on implicit neural feature fields (INFF-Net). It organically combines continuous spatial-angle implicit encoding, importance-guided sparse region-of-interest (RoI) sampling, and ellipse-shape-aware sampling to establish a unified framework that collaboratively addresses the aforementioned issues. As shown in the bottom of Fig. 1, our proposed method improves the baseline architecture from three aspects. The implicit rotation feature field (IRFF) is introduced at the feature extraction stage to address insufficient rotation-aware representation via spatial-angular continuous encoding and a rotation consistency regularization loss. The importance-guided sparse sampling modulation (ISSM) is introduced at the RoI feature extraction stage to suppress background noise via importance-guided sparse sampling. The adaptive shape-aware feature enhancement (ASFE) is introduced at the feature enhancement stage to improve the detection accuracy of slender objects via shape-aware elliptical sampling. Extensive experiments on three benchmark remote sensing object detection datasets, including DOTA-v1.0, DIOR-R, and HRSC2016, demonstrate that the proposed method achieves consistent performance gains over state-of-the-art mainstream approaches.

The core contributions of this work are as follows.

1) We introduce INFF-Net into remote sensing rotated object detection, which achieves rotation-aware continuous encoding under soft rotation and isometric constraints, thereby constructing a spatial-angular continuous feature representation.
2) We design an ISSM strategy that replaces the conventional fixed uniform sampling with foreground-aware adaptive sampling, enhancing discriminative foreground features while suppressing background interference.
3) We develop an ASFE module that employs shape-aware sampling and feature modulation to substantially improve the detection accuracy of objects with extreme ARs.

The remainder of this article is organized as follows. In Section II, we review relevant research closely related to this article. In Section III, we provide a detailed introduction of our proposed method. Section IV puts forward the experimental results and discussion. Finally, Section V concludes this article and makes the future plan.

II. Related Work

A. Rotated Object Detection and Equivariant Representation

Rotated object detection in remote sensing aims to locate and classify ground objects oriented in any direction and serves as a core technology for aerial image interpretation, urban monitoring, and marine surveillance [1], [2], [3]. Compared to horizontal bounding box detection, rotated bounding boxes better align with the actual contours of objects and effectively suppress background interference [15]. Depending on the detection framework, mainstream methods can be categorized into three types: two-stage, single-stage, and Transformer-based end-to-end methods.

Two-stage methods generate candidate regions via a RoI proposal network, followed by refine regression and classification of the candidate objects. RoI Transformer [16] converts horizontal candidate boxes into rotated candidate boxes to improve object localization accuracy. Oriented R-CNN [10] uses midpoint offsets to encode oriented bounding boxes, serving as a classic baseline for two-stage detection. Gliding Vertex [17] expands horizontal boxes into rotated bounding boxes using sliding offsets, effectively addressing the issue of angular periodicity. ALS-Teacher [18] introduces direction-preserving interpolation enhancement and a global rotation loss to suppress noise in the rotation object labels. Although these methods perform well in terms of accuracy, their region proposal and regression mechanisms result in slow inference speeds. Single-stage methods directly regress oriented bounding boxes on feature maps. S^2ANet [19] alleviates the misalignment between anchor boxes and convolutional features by designing a feature alignment module. R^3Det [20] uses a feature refinement module to progressively refine rotated bounding boxes, improving detection accuracy for objects with high ARs. RTMDet [21] and DSSNet [22] optimize the detection process from the perspectives of efficient backbone architecture and dynamic sample selection, respectively. Focal Teacher [23] designs a positioning focal loss adapted to rotated frames to alleviate the hard-sample

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learning problem for rotated objects. Although the single-stage approaches improve inference speed, they still have limitations when handling global context and complex occlusion scenarios. Transformer-based methods eliminate anchors and postprocessing, achieving end-to-end detection through an ensemble prediction mechanism. DETR [24] first introduces the Transformer architecture to object detection, followed by Deformable DETR [25], which significantly accelerates convergence through deformable attention. AO^2-DETR [26] adapts the DETR framework to the rotation detection task. ARS-DETR [27] introduces an aspect-ratio-sensitive attention mechanism to improve adaptability for detecting elongated objects. The O^2-DETR series [28] achieves an excellent balance between detection speed and accuracy by incorporating a real-time end-to-end rotation detection Transformer. Although existing methods have formed a diverse architectural ecosystem, most rely on explicit discrete convolutions to construct feature extraction networks, resulting in insufficient continuity and invariance in rotated feature representation. Additionally, fixed RoI sampling introduces background noise, and the limited modeling capability for objects with extreme ARs constrains the detection performance.

To enhance the adaptability of features to object rotation poses, rotation-aware representation has become a key research direction in this field, aiming to enable the features extracted by the network to transform in accordance with changes in the object's rotation poses. At the network architecture level, ReDet [29] first introduces rotation-invariant networks into remote sensing object detection, achieving hierarchical extraction from rotation-invariant to invariant features. ARC [30] and GRA [31] utilize adaptive rotational convolution and grouped rotational attention mechanisms, respectively, to extract rotationally invariant features in parallel from multiple directions, thereby mitigating the pose sensitivity inherent in traditional convolutions. At the frequency-domain transformation level, FAA [32] leverages the rotational invariance of the Fourier transform to estimate the object's principal orientation in the frequency domain and align features, effectively addressing the issue of directional inconsistency in feature fusion. FRED [33] introduces rotation-invariant constraints in both the backbone network and the detection head, enhancing the consistency of features under rotational transformations. KFIoU [34] employs Gaussian modeling and Kalman filtering to propose a rotation loss function, thereby smoothing angular boundary transitions. Although these methods [29], [30], [31], [32], [33], [34] have made progress in rotated modeling, they remain fundamentally limited by discrete convolution and fixed sampling patterns. Specifically, they only support limited discrete orientations and cannot achieve unified continuous spatial-angle joint encoding for arbitrary-orientation objects. Besides, fixed RoI sampling brings background noise and poorly fits elongated objects. Overall, existing methods are not yet capable of simultaneously addressing the aforementioned limitations.

Different from existing methods, this article proposes INFF-Net, which constructs a unified framework to collaboratively solve these problems by organically integrating continuous spatial-angle implicit encoding, importance-guided sparse RoI sampling, and ellipse-shape-aware sampling.

B. Implicit Neural Representations

Implicit neural representations (INRs) have emerged in recent years as an important research paradigm in computer vision and graphics. The core idea is to employ multilayer perceptrons (MLPs) to map discrete data into continuous features, circumventing the structural limitations of discrete convolutions and achieving continuous and smooth representations. NeRF [35] pioneers a heuristic paradigm of neural rendering by representing scenes as continuous radiance fields through implicit representation. SIREN [36] employs periodic activation functions to enable implicit networks to effectively capture high-frequency details. Multiresolution hash encoding [37] maps coordinates to high-frequency implicit representations via learnable feature grids, achieving rapid fitting of complex signals with constant computational complexity, thereby laying an encoding foundation for introducing implicit representations into detection tasks. In the field of object detection, CFRN [38] represents multiscale features as continuous functions and directly decodes predictions at arbitrary coordinates through a frequency-domain implicit feature decoder. The DeLU activation function [39] improves the applicability of SIREN [36] to implicit representation of remote sensing images, enhancing the stability of continuous coordinate feature decoding in remote sensing scenarios. Yang et al. [40] encode 3-D coordinates into implicit transformation representations of image features, establishing a continuous mapping from 2-D observations to 3-D space.

Although INRs exhibit remarkable advantages in continuous signal modeling, their application to rotated object detection in remote sensing remains to be explored. Existing works primarily toward natural images and 3-D scenes, lacking specialized designs for joint spatial-angular representation of rotated objects. INRs can map discrete coordinates to continuous feature representations, which naturally aligns with the demands of rotated object detection for angular continuity and spatial smoothness, effectively overcoming the inherent pose sensitivity of discrete convolutions.

C. RoI Feature Extraction and Slender Object Modeling

RoI feature extraction is a core component of the two-stage detection framework, aiming to extract fixed-size feature representations from candidate regions, directly determining the purity of foreground features and the final detection accuracy. RoI Align [41] eliminates quantization errors through bilinear interpolation, becoming the standard for RoI feature extraction in horizontal object detection. For rotated object detection, rotated RoI (RRoI) Align [16] extends RoI Align to the rotated manner, enabling accurate feature extraction within rotated bounding boxes. Oriented R-CNN [10] further optimizes feature encoding efficiency by introducing a midpoint offset representation upon RRoI Align. Oriented RepPoints [42] represent rotated objects with adaptive point sets, flexibly capturing the contour features of arbitrarily shaped objects through deformable sampling. BBAVectors [43] decompose a rotated bounding box into boundary vectors from the center point to its four sides, achieving orientation-aware feature extraction within an anchor-free framework. DODet [44]

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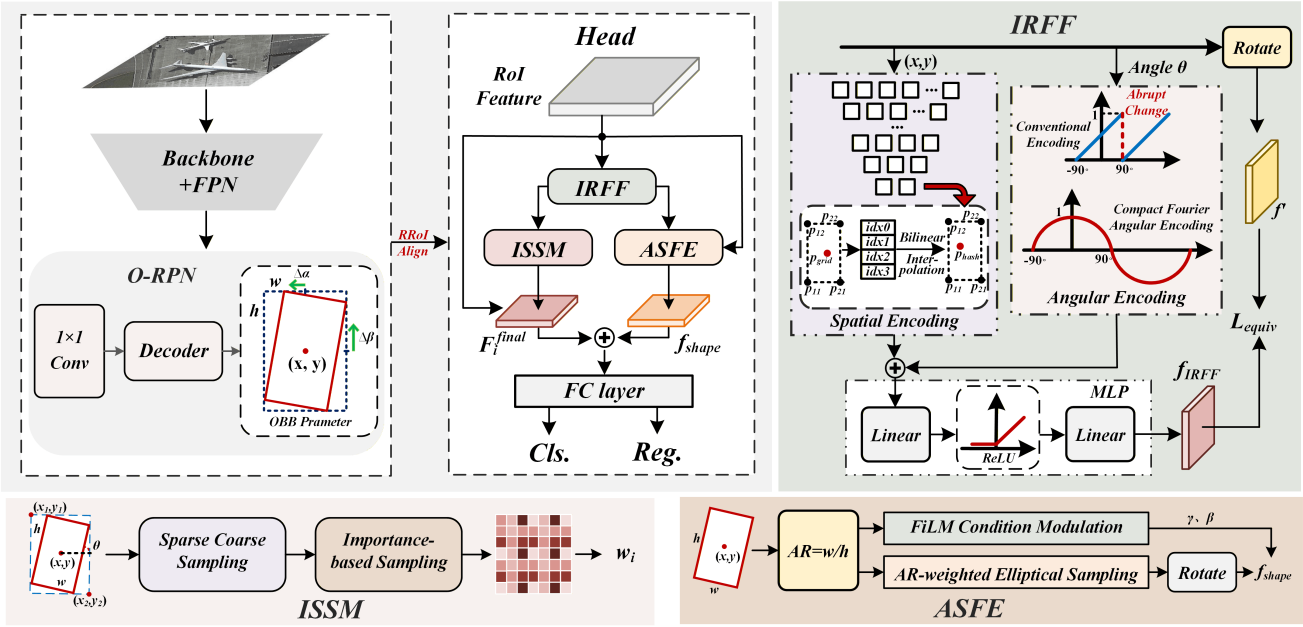


Fig. 2. Overview framework of INFF-Net.

refines RoI features through a dual-alignment mechanism, improving feature extraction accuracy. HA-RDet [45] adopts a hybrid anchor design to enhance RoI quality and incorporates orientation-aware convolution. However, most of the above methods employ uniform sampling strategies with fixed density or distribution, extracting features indiscriminately from all positions within the candidate region. This makes it difficult to dynamically distinguish foreground object regions from background noise, reducing the signal-to-noise ratio and consequently degrading detection performance.

Furthermore, remote sensing images contain a large number of slender objects with extreme ARs, whose arbitrary anisotropic geometry poses unique challenges for feature extraction. To address this issue, most existing studies have explored it from two main perspectives: backbone network design and label assignment strategies. In terms of backbone network designs, Strip R-CNN [46] first introduces large-size strip convolutions to remote sensing object detection, efficiently capturing directional context for slender objects with fewer parameters. LSKNet [47] employs large-kernel convolutions with a spatial selection mechanism, dynamically adjusting the receptive field to accommodate the contextual requirements of objects with different ARs. PKINet [48] constructs a multiscale receptive field using parallel multikernel convolutions and introduces a context anchor attention mechanism to capture long-range dependencies. For label assignment, SADA-Net [49] and ABP-Net [50], respectively, design shape-aware quality assessment and aspect-ratio-sensitive label assignment strategies, improving the boundary perception of slender objects. Wei and Wei [51] propose an aspect-ratio-aware dynamic label assignment strategy for rotated object detection, dynamically selecting positive and negative samples to improve overall positive sample quality. DenseRRSD [52] uses a dense set of representative points to represent ship objects, effectively capturing the objects' local semantic information.

The above methods have improved the detection accuracy of objects with extreme ARs from different perspectives, yet they still predominantly adopt fixed sampling patterns within RoIs, struggling to adaptively match the geometric morphological variations of objects with different ARs, resulting in insufficient extraction of contour features for slender objects. To address the above issues, this article proposes an ASFE module, which dynamically adjusts the number of sampling points based on the AR of the objects and employs an elliptical sampling strategy to ensure that the sampling distribution adaptively matches the geometric shapes of objects with different ARs.

III. Methodology

To address the three major bottlenecks in traditional convolutional neural networks for remote sensing rotated object detection—lacking of rotation-aware representation, poor discrimination ability with redundant background noise, and insufficient modeling performance for objects with extreme ARs, this article proposes INFF-Net, a rotation-aware remote sensing object detection method based on an INFF-Net, using Oriented R-CNN [10] as the base framework. The overall framework is shown in Fig. 2.

Specifically, we first construct an IRFF to establish a spatial-angular joint implicit feature representation that is sensitive to rotation, thereby alleviating the difficulty of characterizing rotated objects. Second, we design an ISSM module. By combining sparse coarse sampling with an importance score-guided fine sampling strategy, we generate attention weights to modulate RoI features, thereby enhancing foreground features while suppressing redundant background information. Finally, we introduce an ASFE module to accurately capture the shape

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features of objects with different ARs, thereby improving the modeling performance for objects with extreme ARs.

A. Implicit Rotation Feature Field

Existing methods rely on discrete convolution kernels with fixed structures for feature representation, making them highly sensitive to rotated objects. This often leads to inconsistent feature representations, which in turn affects detection accuracy. To address this, this method proposes an IRFF based on joint spatial-angular encoding, which independently encodes the position and angle of candidate bounding boxes and then concatenates them. Additionally, a rotation consistency regularization loss is introduced to encourage the network to maintain consistent feature representations under rotation transformations, thereby promoting approximate rotation equivariance. The specific structure is shown in Fig. 2.

First, the input pixel coordinates are normalized to [0, 1], providing standardized input for multiscale feature representation. The normalization process is defined as

$$x_{norm} = \frac{x}{S}, \quad y_{\text{norm}} = \frac{y}{S} \tag{1}$$

where S denotes the fixed spatial size to which all input images are resized (we use $S = 1024$ for DOTA-v1.0 and $S = 800$ for DIOR-R and HRSC2016). x_{norm} and y_{norm} represent the normalized horizontal and vertical coordinates, respectively.

On this basis, a multiresolution hash grid structure is constructed, with the resolution of each grid layer increasing exponentially to cover multiscale objects in remote sensing imagery. The resolution of Layer l is computed as

$$res_l = R_{min} \cdot \left(\frac{R_{max}}{R_{min}}\right)^{\frac{l-1}{L-1}}, \quad l \in 1, 2, \ldots, 16 \tag{2}$$

where l is the hierarchical index of the hash grid; R_{min} is the minimum grid resolution, which is used to capture fine-grained features of small-sized objects; R_{max} is the maximum grid resolution, which is employed to model global features of large-sized objects; and L represents the total number of levels in the hash table.

For each hash grid layer, the normalized coordinates are scaled to the resolution range of the current layer, after which learnable hash tables are queried through hash indexing to obtain vertex features. Subsequently, bilinear interpolation is employed to compute the layer-specific features of the current coordinates, ensuring the continuity of spatial features. Finally, the features from each layer of the hash grid are concatenated to obtain the spatial encoding features.

Concurrently, to achieve continuous encoding of rotation angles and mitigate the inherent boundary discontinuity problem in traditional angle encoding, compact Fourier feature encoding is used to map the input rotation angle. The encoding process is described as follows:

$$\gamma(\theta) = \left[\sin\left(2^k\theta\right), \cos\left(2^k\theta\right)\right]_{k=0}^{K-1} \tag{3}$$

where K denotes the number of Fourier encoding frequencies and $\gamma(\theta)$ represents the output angle encoding features.

Subsequently, the concatenated features of spatial encoding and angular encoding are input into a compact multilayer perceptron (MLP) for feature refinement, achieving deep fusion of spatial and angular features to obtain the implicit features. The feature refinement process is as follows:

$$f_\theta = W_2 \cdot ReLU\left(W_1 \cdot \left[f_{hash}; \gamma(\theta)\right] + b_1\right) + b_2 \tag{4}$$

where $[\cdot\,;\cdot]$ denotes the feature concatenation operation, W_1 and W_2 are the weight matrices for the two layers of the MLP, b_1 and b_2 are the corresponding offsets, and f_{hash} denotes the spatially encoded features obtained by stitching together multiresolution hash grids following bilinear interpolation.

To ensure that the implicit feature field learns rotation-consistent representations, we introduce a rotation consistency regularization loss during training. First, we define a 2-D rotation matrix $R(\alpha)$ to apply a rotation transformation to the coordinates and angles and compute the implicit features f' of the rotated input. At the same time, we reshape the original features f into a 2-D vector along the channel dimension and apply a rotation transformation using this rotation matrix to align the spatial structure roughly preserved in the channel dimension, thereby enabling us to measure the consistency of the features before and after rotation. It should be noted that this is not an exact geometric transformation of the feature space, but rather an approximate alignment strategy designed for loss calculation. The rotation matrix and loss function are formulated as follows:

$$R(\alpha) = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}, \quad \alpha \in [0, \pi) \tag{5}$$

$$L_{equiv} = mse\left(f', R(\alpha) \cdot f\right) \tag{6}$$

where f denotes the implicit feature of the original input and f' represents the implicit feature of the rotated input, i.e., the implicit feature obtained by performing spatial rotation transformation on the coordinates and angles via the rotation matrix. $R(\alpha)$ denotes the rotation matrix. Since the rotated bounding boxes in the remote sensing reference data exhibit periodicity (where angles are typically defined using π as a unit of measurement), the angle range is restricted to $\alpha \in [0, \pi)$. Finally, the total training loss is defined as

$$L_{total} = L_{det} + \lambda L_{equiv} \tag{7}$$

where L_{det} denotes the primary task loss (including classification and regression) and λ represents the rotation-equivariance loss weight.

IRFF effectively alleviates the pose-sensitivity issue inherent in discrete convolutions through continuous spatial-angular joint encoding and the rotation consistency regularization, providing robust feature representations for rotated objects. However, IRFF still uses uniform sampling inside RoIs, which tends to introduce background noise. To address this limitation, we propose the ISSM module, effectively suppressing background interference.

B. Importance-Guided Sparse Sampling Modulation

Most existing methods for extracting features from RRoIs employ a uniform sampling strategy with fixed densities,

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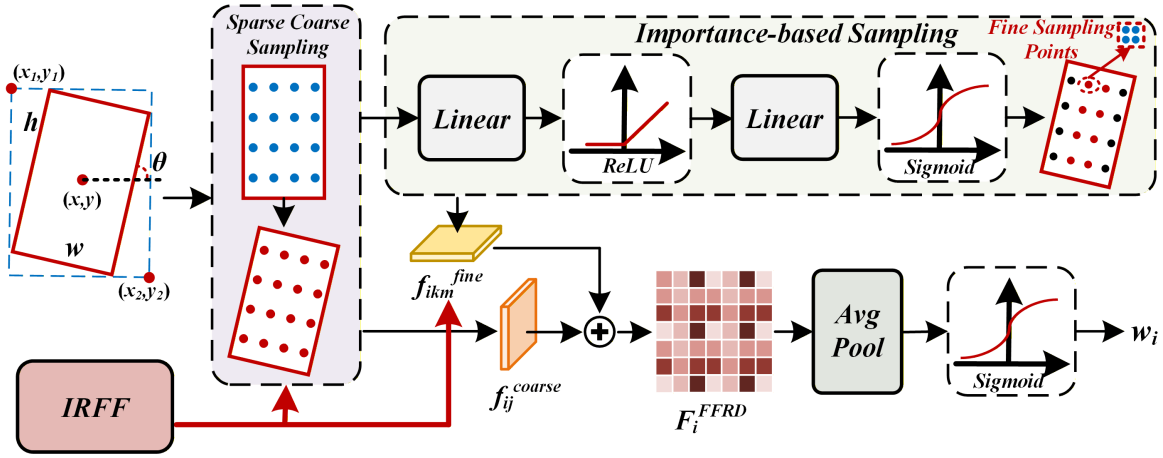


Fig. 3. Design details of ISSM module.

which tends to introduce a significant amount of background noise and reduced detection accuracy. To address these issues, this method proposes an ISSM module, whose structure is shown in Fig. 3.

Specifically, when performing RoI feature extraction, analysis of geometric parameter is first conducted on each rotated candidate box within each batch to obtain its circumscribed horizontal rectangle coordinates, center coordinates, width, height, and rotation angle, providing the geometric foundation for subsequent sampling operations. Analysis of the parameter process is as follows:

$$\mathcal{B}_i = \left(x_1^i, y_1^i, x_2^i, y_2^i, \theta_i\right), \begin{cases} \left(x_c^i, y_c^i\right) = \left(\dfrac{x_1^i + x_2^i}{2}, \dfrac{y_1^i + y_2^i}{2}\right) \\ w_i = x_2^i - x_1^i, h_i = y_2^i - y_1^i \end{cases} \tag{8}$$

where (x_1^i, y_1^i) and (x_2^i, y_2^i) are the pixel coordinates of the upper left and lower right corners of the circumscribed horizontal rectangle of the ith candidate box, respectively; (x_c^i, y_c^i) denotes the center coordinates of the circumscribed horizontal rectangle; and w_i and h_i denote the width and height of the circumscribed horizontal rectangle, respectively.

On this basis, normalized grid offsets are generated in the local coordinate system of the candidate box; element-wise multiplication is applied to these offsets with the candidate box width and height to obtain local sampling coordinates, after which pose alignment is achieved through the rotation matrix, and the coordinates are finally transformed to the global image coordinate system to obtain the global pixel coordinates of sparse sampling points. The coordinate generation process for sparse sampling points is expressed as follows:

$$p_{ij}^{coarse} = R(\alpha) \cdot \left(\Delta_j \odot \begin{bmatrix} w_i \\ h_i \end{bmatrix}\right) + \begin{bmatrix} x_c^i \\ y_c^i \end{bmatrix}, \quad j = 1, 2, \ldots, G^2 \tag{9}$$

where i denotes the index of the ith candidate bounding box in the batch; G is the size of the coarse sampling grid; Δ_j is the normalized grid offset of the jth sampling point, which is generated from uniformly distributed grid coordinates; and $\odot$ denotes the element-wise multiplication operation.

After inputting the coordinates of all coarse sampling points and the corresponding candidate box rotation angles into the IRFF to obtain coarse sampling features, the model predicts the foreground confidence score for each coarse sampling grid based on the importance score, thereby distinguishing the foreground from the background. The computation of importance scores proceeds as follows:

$$f_{ij}^{coarse} = f_\phi\left(x_{ij}^{coarse}, y_{ij}^{coarse}, \theta_i\right) \tag{10}$$

$$s_{ij} = \sigma\left(W_s \cdot f_{ij}^{coarse} + b_s\right) \tag{11}$$

where $(x_{ij}^{coarse}, y_{ij}^{coarse})$ represents the coordinate components of coarse sampling points, f_{ij}^{coarse} is the implicit feature of the corresponding sampling point, $\sigma(\cdot)$ is the sigmoid activation function, and W_s and b_s are the learnable parameters of the importance predictor. s_{ij} is the importance score of the jth coarse sampling grid in the ith candidate box, with higher scores indicating larger confidence scores that the region belongs to foreground objects.

Based on the importance scores, we select high-scoring grids that exceed the threshold τ within each candidate box. For each selected grid, we uniformly sample fine points, feed their coordinates and rotation angles into IRFF to extract features, and apply average pooling within each grid to obtain grid-level fine sampling features. The extraction process of fine sampling features is as follows:

$$f_{ikm}^{fine} = f_\phi\left(x_{ikm}^{fine}, y_{ikm}^{fine}, \theta_i\right) \tag{12}$$

where i stands for the candidate box index, k denotes the index of the high-score grid, m represents the index of the fine sampling point within the grid, and f_{ikm}^{fine} denotes the implicit features of the corresponding fine sampling point.

Average pooling is applied to the fine sampling features within each high-scoring grid to obtain grid-level fine sampling features, which are then fused with coarse sampling features. The fused features are ultimately reshaped into a spatial feature map, with the feature reconstruction process as follows:

$$F_i = \frac{1}{G^2}\sum_{j=1}^{G^2} f_{ij}^{coarse} + \frac{1}{K}\sum_{k=1}^{K}\frac{1}{S'}\sum_{m=1}^{S'} f_{ikm}^{fine} \tag{13}$$

$$F_i^{FFRD} = Interpolate\left(Reshape\left(F_i, (C, G, G)\right)\right) \tag{14}$$

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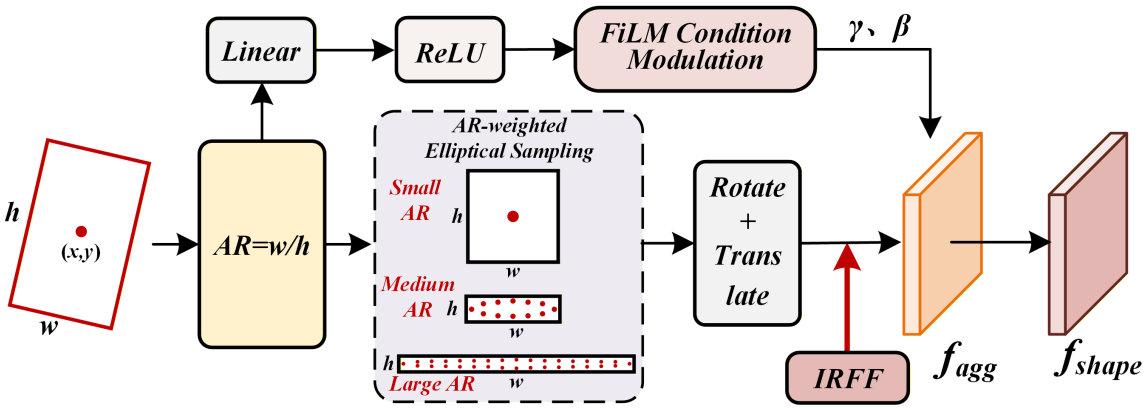


Fig. 4. Design details of ASFE module.

where K is the number of high-score grids for candidate boxes, S' denotes the number of fine sampling points within each high-scoring grid, and G represents the coarse sampling grid size. C is the feature channel dimension, $Reshape(\cdot)$ denotes the feature reshape operation, $Interpolate(\cdot)$ denotes the bilinear interpolation operation, and F_i^{FFRD} represents the ultimately output rotation-aware RoI feature map.

To further suppress background noise and strengthen the feature response of foreground objects, channel attention weights are generated based on the output feature map to modulate the base features. The modulation process is as follows:

$$w_i = \sigma\left(AvgPool\left(F_i^{FFRD}\right)\right), \quad F_i^{final} = w_i \odot F_i^{base} \tag{15}$$

where $AvgPool(\cdot)$ is the average pooling operation, w_i is the generated channel attention weight, F_i^{base} is the RoI features extracted by RoI Align, $\odot$ denotes the element-wise multiplication operation, and F_i^{final} is the final enhanced RoI feature.

ISSM effectively suppresses background noise by concentrating sampling points on high-response regions through coarse-to-fine sampling guided by importance scores. However, its regular-grid-based sampling pattern remains agnostic to object shape. For slender objects with extreme ARs, fixed grids cannot align with their elongated contours, causing a considerable number of sampling points to deviate from foreground regions. We thus propose ASFE, which dynamically adapts to slender geometries via shape-aware elliptical sampling.

C. Adaptive Shape-Aware Feature Enhancement

Many rotated objects in remote sensing images have extremely high ARs, making it difficult for fixed sampling patterns to effectively capture their shape features, which can easily lead to missed detections and false positives. To address this limitation, we propose an ASFE module, whose detailed structural schematic is illustrated in Fig. 4.

Specifically, when performing feature extraction on rotated candidate boxes, geometric decomposition is first executed on the input rotated candidate box to compute its width (w), height (h), and AR. Subsequently, based on the specific value of AR, the candidate boxes are categorized into three groups, dynamically determining the number of sampling points for each candidate box to capture the shape features of objects with different ARs. The specific computation of sampling points is as follows:

$$N_{ex} = \min\left(\max\left(\lceil AR \rceil, N_{base}\right), N_{max}\right) \tag{16}$$

$$N_{sample} = \begin{cases} 1, & AR < T_{AR} \\ N_{base}, & T_{AR} \le AR < T_{ext} \\ N_{ex}, & AR \ge T_{ext} \end{cases} \tag{17}$$

where $\lceil \cdot \rceil$ stands for ceiling operator, T_{AR} denotes the basic grading threshold for the AR, T_{ext} represents the extreme AR threshold, N_{base} stands for the basic number of sampling points, and N_{max} is the maximum number of sampling points.

On this basis, with the candidate box center as the origin, uniformly distributed elliptical sampling points are generated in the local coordinate system, then rotationally transformed through the rotation matrix, and translated to the global image coordinate system, ensuring strict alignment between the sampling points and the rotational pose of the object. The transformation process from local to global coordinates is as follows:

$$p_{local}^m = \left[\frac{w}{2} \cdot \cos\phi_m, \frac{h}{2} \cdot \sin\phi_m\right]^{\mathrm{T}}, \quad m = 1, 2, \ldots, N_{sample} \tag{18}$$

$$p_{global}^m = R(\theta) \cdot p_{local}^m + \begin{bmatrix} x_c \\ y_c \end{bmatrix} \tag{19}$$

where ϕ_m is the polar angle of the mth sampling point, (x_c, y_c) denotes the center coordinates of the candidate box, θ represents the rotation angle of the candidate box, and $R(\theta)$ is the rotation matrix.

The feature extraction process for all sampling points fully shares the IRFF backbone network, ensuring the rotation equivariance and continuity of features. After obtaining the features of all sampling points, aggregated features at the candidate box level are obtained based on the batch indices of the candidate boxes. The aggregation process is as follows:

$$f_{agg} = \frac{1}{N_{sample}} \sum_{m=1}^{N_{sample}} f_\phi\left(x^m, y^m, \theta\right) \tag{20}$$

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where (x^m, y^m) represents the global coordinate of the mth sampling point and f_{agg} denotes the aggregated candidate box-level features.

To further enhance the feature discriminability for objects with different ARs, the AR is first logarithmically transformed, after which two independent linear layers generate scaling coefficients γ and bias coefficients β, performing element-wise linear modulation on the aggregated features to achieve shape-aware feature enhancement. The modulation process is as follows:

$$h_i = ReLU\left(W_h \cdot \log\left(AR + \epsilon\right) + b_h\right) \tag{21}$$

$$f_{shape} = \gamma \odot f_{agg} + \beta, \begin{cases} \gamma = W_\gamma \cdot h_i + b_\gamma \\ \beta = W_\beta \cdot h_i + b_\gamma \end{cases} \tag{22}$$

where ϵ is the extremely small positive number (we set 10^{-6} in this article), which is used to ensure numerical stability and prevent logarithmic operations from tending toward negative infinity when the AR is extremely small. $\odot$ denotes the element-wise multiplication operation, and W_h and b_h are the learnable parameters. W_γ, b_γ, W_β, and b_β are the learnable parameters of the modulation parameter generation head, and f_{shape} is the final output shape-aware enhanced features.

IV. Experiments

A. Experimental Setup

1) Datasets: To comprehensively evaluate the performance and generalization capabilities of the method proposed in this article, experiments are conducted on three widely used datasets for remote sensing rotated object detection: DOTA-v1.0, DIOR-R, and HRSC2016.

DOTA-v1.0 [53] comprises 2806 aerial images and 188 282 instances, covering 15 common land cover classes. Image resolutions range from 800×800 to 4000×4000 pixels, and the objects exhibit characteristics such as arbitrary detections, large-scale variations, and dense distribution.

DIOR-R [54] is a rotation-annotated extension of the DIOR dataset, comprising 23 463 optical remote sensing images and 192 472 rotation-annotated instances, covering 20 land cover classes. All images are standardized to a resolution of 800×800 pixels and are characterized by high interclass similarity and complex background interference.

HRSC2016 [55] is a dataset designed for ship detection, comprising 1061 high-resolution aerial images and 2976 ship instances. Image dimensions range from 300×300 to 1500×900 pixels, and the ship objects exhibit characteristics such as extreme ARs, arbitrary orientations, and dense arrangements.

2) Evaluation Metrics: In accordance with standard evaluation protocols in this field, this article uses the mean average precision (mAP) as the primary quantitative metric. On the DOTA-v1.0 and DIOR-R datasets, the VOC2007 evaluation standard is adopted (with an IoU threshold of 0.5 for calculating mAP). On the HRSC2016 dataset, both VOC2007 and VOC2012 metrics are reported to provide a more comprehensive assessment of localization accuracy.

TABLE I

Module Ablation Study on the DOTA Dataset. All Results Are Obtained Under the Same Experimental Settings. The Baseline Refers to Our Reimplemented Oriented R-CNN [10], Which May Differ From the Originally Reported Value Due to Implementation Details

Datasets	Modules			mAP(%)
	IRFF	ISSM	ASFE	
DOTA	-	-	-	73.95
	✓	-	-	75.79
	-	✓	-	75.36
	-	-	✓	74.91
	✓	✓	-	76.80
	✓	-	✓	76.53
	-	✓	✓	76.64
	✓	✓	✓	**77.87**

3) Implementation Details: The method described in this article is implemented using the MMRotate [56] platform, with Oriented R-CNN [10] serving as the base architecture and ResNet-50 as the backbone network. Data augmentation includes random horizontal flipping, vertical flipping, and random rotation. The model is trained for 36 epochs on the DOTA-v1.0 and DIOR-R datasets and 72 epochs on the HRSC2016 dataset to ensure full convergence. All experiments are conducted on an NVIDIA RTX 4090 with 24 GB GPU using the PyTorch framework. During training, the SGD optimizer is employed with a momentum of 0.9, a weight decay of 0.0001, and a batch size of 32. The initial learning rates are set to 0.005, 0.01, and 0.005 for the DOTA-v1.0, DIOR-R, and HRSC2016 datasets, respectively. The input image sizes are fixed to 1024×1024 pixels for DOTA-v1.0 and 800×800 pixels for DIOR-R and HRSC2016.

B. Ablation Study

1) Module Ablation Study: To quantify the contributions of each module, this article uses Oriented R-CNN [10] as the baseline model and conducts module ablation experiments on DOTA-v1.0. The results are shown in Table I.

As shown in Table I, the baseline model, which lacks rotation invariance due to standard convolution and uses uniformly distributed sampling points, achieves only 73.95% mAP on DOTA-v1.0. After introducing IRFF, the mAP improves by 1.84%, validating the effectiveness of rotation-aware feature extraction. When ISSM was applied alone, the mAP reached 75.36%, indicating that ISSM possesses an independent capability to suppress background interference. Similarly, applying ASFE alone achieved an mAP of 74.91%, validating the independent benefits of elliptical shape-aware sampling for modeling elongated objects. Pairwise combinations further reveal the complementary nature of the modules. The combination of IRFF and ISSM achieved an mAP of 76.80%, indicating that the continuous rotation feature basis provided by IRFF enables ISSM to distinguish between foreground and background more effectively. The combination of IRFF and ASFE achieved an mAP of 76.53%, demonstrating that shape-adaptive elliptical sampling benefits from the continuous rotation feature field and can more precisely fit the

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TABLE II
ABLATION STUDY OF IRFF PARAMETERS ON DOTA-V1.0

IRFF				
L	mAP(%)		K	mAP(%)
8	76.22		2	75.78
16	**77.87**		4	**77.87**
24	76.83		8	76.56
λ	0	0.1	0.5	1.0
mAP(%)	76.40	76.79	**77.87**	76.97

object's contours. Furthermore, when ISSM is combined with ASFE, the mAP reaches 76.64%, confirming that even in the absence of continuous rotation encoding, the synergy between importance-guided sampling and shape-aware sampling still yields performance gains. When all three modules operate in concert, the model achieves an mAP of 77.87% on DOTA-v1.0 (a 3.92% improvement over the baseline), fully validating the complementary strengths of IRFF, ISSM, and ASFE in rotation-aware representation, background suppression, and shape adaptation.

2) IRFF Ablation Study: The performance of the IRFF module is jointly influenced by the hash grid level L, the Fourier coding frequency K, and the rotation-invariant loss weight λ. This article conducts ablation experiments on DOTA-v1.0, and the results are shown in Table II.

First, according to the analysis by Müller et al. [37], 2-D visual signals can effectively cover the spectrum from local to global scales within 8–16 levels. Considering the large-scale variations in remote sensing imagery, this method appropriately expands the search range. Experimental results indicate that the model's detection performance follows a trend of initially increasing and then decreasing with the number of levels. When $L = 8$, the insufficient number of levels leads to inadequate spatial representation. When $L = 24$, an excessive number of levels introduces redundant high-frequency features. To balance multiscale representation capability and model generalization, we select $L = 16$.

Second, based on the theoretical analysis of Fourier feature coding by Tancik et al. [57], the number of frequencies should match the spectral characteristics of the objective function. Too few frequencies result in insufficient angular discrimination, while too many frequencies cause spectral aliasing, which can lead to overfitting. Experimental results indicate that optimal performance is achieved when $K = 4$; when $K = 2$, spectral poverty leads to homogenization of angular features, resulting in reduced accuracy; and when $K = 8$, the model is prone to overfitting in high-frequency components.

Finally, the rotation consistency regularization loss weight λ controls the strength of the geometric constraints relative to the main detection task. Experimental results show that when $\lambda = 0$, the lack of isotropic constraints leads to the lowest accuracy. When $\lambda = 0.1$, the weak constraints provide some improvement, but geometric consistency remains insufficient, resulting in only a moderate gain. Meanwhile, when $\lambda = 1.0$, excess constraints suppress learning for both classification and regression, and when $\lambda = 0.5$, an optimal balance is achieved between constraint strength and flexibility for the main task.

TABLE III
ABLATION STUDY OF ISSM PARAMETERS ON DOTA-V1.0

ISSM			
G	mAP(%)	τ	mAP(%)
2	76.79	0.3	77.06
4	**77.87**	0.5	**77.87**
8	77.05	0.7	77.17

TABLE IV
ABLATION STUDY OF ASFE PARAMETERS ON DOTA-V1.0

ASFE					
N_{base}	mAP(%)	T_{AR}	mAP(%)	T_{ext}	mAP(%)
8	76.45	3.0	77.10	10.0	77.21
12	**77.87**	5.0	**77.87**	15.0	**77.87**
16	76.53	8.0	76.52	20.0	76.91

3) ISSM Ablation Study: To validate the validity of sparse sampling and importance filtering, this article conducts an ablation study on the coarse sampling grid size G and the importance filtering threshold τ. The results are shown in Table III.

Based on the typical scale of remote sensing objects in the RoI feature map, small objects typically occupy approximately from 4×4 to 8×8 pixels. Therefore, candidate values are selected centered on $G = 4$ and extending to both sides. Experimental results indicate that when $G = 2$, the sampling coverage is too sparse, making it difficult to accurately locate the foreground regions. When $G = 8$, the grid coverage is too dense, which is not conducive to robust classification. When $G = 4$, optimal accuracy is achieved.

The importance threshold τ controls the strictness of filtering high-resolution grids from the coarse sampling grid into the fine sampling; 0.5 serves as a natural decision boundary and has been validated as a robust central threshold in dynamic sample screening strategies [49], [51]. Experimental results show that $\tau = 0.3$ retains too many background grid cells, $\tau = 0.7$ is too strict, discarding some foreground grid cells containing object edge information, and $\tau = 0.5$ achieves the optimal balance between recall and precision.

4) ASFE Ablation Study: To validate the effectiveness of AR threshold and elliptical sampling, this article conducts a parameter study on the base number of sampling points (N_{base}), the AR binning threshold (T_{AR}), and the extreme AR threshold (T_{ext}). The results are shown in Table IV.

RRoI Align [16] figures that the sampling efficiency of an elliptical grid is approximately four times that of a rectangular grid. Setting N_{base} to around 12 ensures effective coverage, with upper and lower bounds selected from candidate values. The ARs of remote sensing objects are concentrated around 5:1 [50], while those of extreme objects are concentrated in the range of 12:1–15:1 [54], [55]. Experimental results indicate that optimal performance is achieved when $N_{base} = 12$, $T_{AR} = 5$, and $T_{ext} = 15$. When $N_{base} = 8$, the number of sampling points is insufficient, making it difficult to fully capture the contour features of elongated objects.

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TABLE V

PERFORMANCE COMPARISONS ON DOTA-V1.0. CATEGORY ABBREVIATIONS ARE DEFINED AS FOLLOWS: PLANE (PL), BASEBALL DIAMOND (BD), BRIDGE (BR), GROUND TRACK FIELD (GTF), SMALL VEHICLE (SV), LARGE VEHICLE (LV), SHIP (SH), TENNIS COURT (TC), BASKETBALL COURT (BC), STORAGE TANK (ST), SOCCER BALL FIELD (SBF), ROUNDABOUT (RA), HARBOR (HA), SWIMMING POOL (SP), AND HELICOPTER (HC). WE BOLD THE BEST RESULTS AND UNDERLINE THE SECOND-BEST RESULTS

Methods	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP(%)
RetinaNet-O [13]	89.41	76.83	40.90	67.57	77.51	62.67	77.54	90.89	82.34	81.99	58.16	61.56	56.46	63.70	38.96	68.43
RoI Transformer [16]	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
PSCD [58]	89.32	82.29	37.92	71.52	78.40	66.33	78.01	90.89	84.21	80.63	60.22	64.73	59.69	68.37	53.85	71.09
CenterMap-Net [59]	88.88	81.24	53.15	60.65	78.62	66.55	78.10	88.83	77.80	83.16	49.36	66.19	72.10	72.36	58.70	71.74
ILOEDet [60]	88.74	76.00	50.84	63.00	79.36	75.45	86.93	90.90	85.69	85.13	64.12	65.33	65.84	70.45	44.43	72.81
GRDet [61]	**89.90**	81.21	52.03	66.48	71.45	70.60	74.80	90.85	**87.96**	81.32	65.08	66.70	63.63	69.04	65.45	73.10
Rotated Faster R-CNN [9]	89.40	81.81	47.28	67.44	73.96	73.12	85.03	90.90	85.15	84.90	56.60	64.77	64.70	70.28	62.22	73.17
S^2ANet [19]	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12
HRDet [62]	88.96	81.90	50.47	68.23	79.93	80.41	87.33	90.90	81.74	84.10	56.00	61.63	73.30	73.45	59.21	74.50
G.V. [17]	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32	75.02
RSDet++ [63]	86.80	82.70	54.60	71.70	76.00	71.20	83.50	87.40	83.40	85.30	**72.40**	62.90	70.90	72.30	**70.40**	75.40
O-RCNN [10]	89.46	82.12	54.78	70.86	78.93	83.00	88.20	90.90	87.50	84.68	63.97	67.69	74.94	68.84	52.28	75.87
O-RepPoints [42]	87.02	83.17	54.13	71.16	80.18	78.40	87.28	90.90	85.97	86.25	59.90	70.49	73.53	72.27	58.97	75.97
DGRL [2]	89.04	83.87	52.01	73.46	78.69	83.27	88.38	90.90	87.07	84.86	65.04	65.72	75.04	67.98	57.26	76.17
ReDet [29]	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
SREDet [64]	89.36	85.51	50.87	74.52	80.50	74.78	86.43	90.01	87.40	83.97	64.36	69.10	67.72	73.65	65.93	76.34
FAA [32]	89.40	84.15	54.66	72.23	78.42	82.89	87.86	90.89	86.26	85.24	63.95	65.58	75.30	70.44	61.06	76.55
UFDet [65]	89.40	83.63	53.03	72.81	78.98	81.44	87.91	90.90	87.00	84.40	64.21	69.10	70.87	71.33	63.61	76.57
CTRP [66]	89.09	83.38	53.26	76.11	78.24	83.36	88.34	**90.91**	86.14	84.91	65.94	67.17	76.79	70.86	63.07	77.17
ARC [30]	89.40	82.48	55.33	73.88	79.37	84.05	88.06	90.90	86.44	84.83	63.63	70.32	74.29	71.91	65.43	77.35
LSKNet-S [47]	89.66	**85.52**	**57.72**	75.70	74.95	78.69	88.24	90.88	86.79	86.38	66.92	63.77	77.77	74.47	64.82	77.49
O^2-DFINE [28]	89.08	80.18	55.17	69.73	**80.53**	83.77	88.46	90.86	87.55	**86.86**	60.14	68.68	77.17	77.79	70.03	77.73
Ours	89.30	76.20	54.90	**79.90**	76.10	**84.80**	**89.10**	90.40	75.50	78.80	**72.40**	**72.80**	**78.40**	**86.10**	63.50	**77.87**

When N_{base} = 16, although more points are used, the gain plateaus and even slightly declines due to potential overfitting or redundant sampling. When N_{base} = 12, the best performance (77.87%) is achieved. When T_{AR} = 3, sampling points increase prematurely. When T_{AR} = 8, recognition of elongated objects is delayed, and T_{AR} = 5 precisely matches the natural boundary of object ARs in the dataset. T_{ext} = 10 leads to overallocation of sampling resources, while T_{ext} = 20 is insufficient to cover extreme objects. T_{ext} = 15 achieves the optimal balance. This configuration enables reasonable allocation of sampling resources based on object ARs. Conventional objects use a small number of sampling points to ensure efficiency, while objects with extreme ARs use additional sampling points to capture their complete contours. At the same time, elliptical sampling ensures that the distribution of sampling points better matches the true shape of rotated objects, significantly improving the localization accuracy of elongated objects.

C. Main Results

1) Experimental Results and Analysis on DOTA-v1.0: Table V presents the comparison results on the DOTA-v1.0 dataset. Our method achieves the highest mAP of 77.87% and outperforms other methods in multiple categories with high ARs, including ship (SH) and harbor (HA). It is worth noting that although our method achieves the best overall performance, the bridge (BR) category (54.90%) is slightly lower than LSKNet [47] (57.72%). This may be mainly because bridge objects, as typical slender structures spanning rivers or roads, require large-range cross-domain contextual associations, where large-kernel convolutions have inherent advantages. Nevertheless, our method still outperforms most comparison methods on the BR category and achieves significant improvements over the baseline in other high AR categories, which fully validates the effectiveness of our method for extreme AR objects. O^2-DFINE [28] achieves an mAP of 77.73%, and it employs a real-time end-to-end rotation detection Transformer that effectively balances detection speed and accuracy. LSKNet-S [47] achieves an mAP of 77.49% by dynamically adjusting the receptive field through large-kernel convolutions and a spatial selection mechanism to adapt to objects with different ARs. However, these methods still rely on explicit discrete convolutions and lack constraints for continuous rotation, and they employ a uniform sampling strategy with fixed density, which inevitably introduces background noise. Our method integrates IRFF with ISSM, achieving 0.14% improvements over O^2-DFINE and 0.38% improvements over LSKNet-S.

Our method exhibits a decrease in accuracy for categories such as baseball diamonds (BDs), basketball courts (BCs), and storage tanks (STs). We observe that these objects are highly geometrically symmetrical with limited pose variations within the dataset. As a result, the benefits brought by spatial-angular continuous encoding are inherently limited. Additionally, the ASFE module is optimized for elongated rotated objects, which introduces a slight sampling-distribution mismatch when handling compact square-shaped objects. It should be emphasized, however, that such negative effects mainly manifest for highly symmetric objects with few orientation changes. For compact square-like objects with diverse rotation poses, our modules can still yield noticeable performance gains.

Although the overall improvement relative to some competing methods is limited, our method demonstrates clear advantages for two groups of rotated objects: objects with large

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TABLE VI

PERFORMANCE COMPARISONS ON DIOR-R. CATEGORY ABBREVIATIONS ARE DEFINED AS FOLLOWS: AIRPLANE (APL), AIRPORT (APO), BASEBALL FIELD (BF), BASKETBALL COURT (BC), BRIDGE (BR), CHIMNEY (CH), DAM (DAM), EXPRESSWAY TOLL STATION (ETS), EXPRESSWAY SERVICE AREA (ESA), GOLF FIELD (GF), GROUND TRACK FIELD (GTF), HARBOR (HA), OVERPASS (OP), SHIP (SH), STADIUM (STA), STORAGE TANK (STO), TENNIS COURT (TC), TRAIN STATION (TS), VEHICLE (VE), AND WINDMILL (WM). WE BOLD THE BEST RESULTS AND UNDERLINE THE SECOND-BEST RESULTS

Methods	APL	APO	BF	BC	BR	CH	DAM	ETS	ESA	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	mAP(%)
RetinaNet-O [13]	61.49	28.52	73.57	81.17	23.98	72.54	19.94	72.39	58.20	69.25	79.54	32.14	44.87	77.71	65.57	61.09	81.46	47.33	38.01	60.24	57.55
TIOE-Det [67]	68.65	28.62	76.68	76.87	39.32	74.11	20.36	63.87	72.66	75.19	77.41	40.63	47.48	82.61	72.58	70.33	81.93	47.86	**52.06**	64.24	61.98
DFDet [68]	61.92	38.83	77.41	71.36	34.11	74.97	26.26	62.31	76.06	75.56	79.62	38.26	52.76	80.40	73.11	68.27	81.38	52.23	51.10	64.35	62.11
RFE-FCOS [69]	67.78	33.21	79.80	84.27	36.54	75.82	24.61	76.19	67.54	69.96	83.46	43.99	50.52	81.65	68.19	68.47	83.39	51.13	39.55	60.87	62.35
HRDet [62]	70.17	30.72	70.70	86.45	37.76	71.62	24.76	77.19	63.93	67.02	81.67	42.72	53.61	80.15	75.34	70.35	80.51	56.02	47.34	63.63	62.58
S^2ANet [19]	65.40	42.04	75.15	78.11	36.01	72.61	28.01	65.09	60.36	65.26	80.47	35.91	52.10	82.33	65.89	66.08	84.61	54.13	41.58	69.67	62.90
OriMamba [70]	72.08	35.74	**80.55**	81.30	36.73	72.61	32.62	65.48	79.65	78.11	**84.06**	43.65	50.04	72.25	**81.61**	70.01	81.55	**64.70**	41.58	65.73	64.53
O-RCNN [10]	62.00	44.92	71.78	87.93	43.84	72.64	**35.46**	66.39	81.35	74.10	80.95	43.52	58.42	81.25	68.01	65.52	88.62	59.31	43.27	66.31	64.63
RoI Transformer [16]	63.10	47.19	71.97	81.20	42.49	72.59	30.37	71.08	79.43	**78.15**	82.45	**48.77**	57.82	81.24	76.82	62.48	81.45	57.56	43.80	66.32	64.81
DODet [44]	63.40	43.35	72.11	81.32	43.12	72.59	33.32	**78.77**	70.84	74.15	75.47	48.00	59.31	**85.41**	74.04	71.56	81.52	55.47	51.86	66.40	65.10
ARS-DETR [27]	68.00	**54.17**	74.43	81.65	41.13	75.66	34.89	73.07	**81.92**	76.10	78.62	36.33	55.41	84.55	70.09	**72.23**	81.14	61.52	50.57	**70.28**	66.12
O-RepPoints [42]	67.80	48.01	77.02	85.37	38.55	**78.45**	33.67	72.06	81.13	76.00	79.89	45.72	54.27	85.13	76.04	65.27	85.38	59.76	48.02	68.92	66.31
Ours	**80.80**	44.50	79.30	**89.80**	**46.00**	72.30	30.30	69.40	71.10	76.00	79.40	47.80	**60.10**	81.00	81.10	70.30	**90.10**	59.40	43.70	65.50	**66.67**

ARs and near-square ground objects exhibiting diverse arbitrary orientations. Specifically, we achieve significant improvements across several categories with practical applications: soccer ball fields (SBFs; 72.40%, outperforming LSKNet-S by 5.48% and O^2-DFINE by 12.26%), swimming pools (SPs; 86.10%, outperforming LSKNet-S by 11.63% and O^2-DFINE by 8.31%), large vehicles (LVs; 84.80%, outperforming LSKNet-S by 6.11%), harbors (78.40%), and ships (89.10%), among others. In addition, our method achieves significant improvements over the baseline in categories such as helicopters (+11.22%), soccer ball fields (+8.43%), and harbors (+3.46%). The categories mentioned above include not only objects with high ARs, such as ships, helicopters, and harbors, but also nearly square features with varied orientations, such as SBFs and SPs. These performance improvements were achieved through a combination of spatial-angular continuous coding and elliptical shape perception sampling. For LVs (+1.80%) and SHs (+0.90%), our method consistently outperformed the baseline as well, contributing to the overall mAP gain of +3.92% over the baseline, demonstrating the robustness and generalization capability of the proposed method.

2) Experimental Results and Analysis on DIOR-R: Table VI presents the comparison results on the DIOR-R dataset. Our method achieves an mAP of 66.67% and obtains the best performance across multiple categories, including bridge (BR), overpass (OP), and train station (TS). ARS-DETR [27], which introduces an aspect-ratio-sensitive attention mechanism, achieves an mAP of 66.12%, while O-RepPoints [42], which employs an adaptive point set representation to flexibly capture the contour features of arbitrarily shaped objects, achieves an mAP of 66.31%. However, these methods still rely on fixed sampling strategies and lack explicit rotation constraints, leading to performance bottlenecks in complex scenes with diverse orientations and dense arrangements. Our method enhances performance by integrating an IRFF with adaptive shape-aware features, achieving 0.36% improvements over O-RepPoints and 0.55% improvements over ARS-DETR.

3) Experimental Results and Analysis on HRSC2016: Table VII presents the comparison results on the HRSC2016 dataset. Our method achieves an mAP of 90.50% under the VOC2007 metrics and 97.90% under the VOC2012 metrics. DynaPro [73] employs adaptive-constrained dynamic progressive activation for rotated object detection and achieves second-best performance under the VOC2012 metric. DSSNet [22] optimizes the anchor-free rotated object detection process through dynamic sample selection, achieving mAPs of 90.38% (VOC2007) and 97.26% (VOC2012). However, these methods still rely on explicit discrete convolution for feature extraction and lack constraints such as continuous rotation, resulting in insufficient stability of feature representation under different rotational poses. The method proposed in this article effectively mitigates these issues through the synergistic interaction of three major modules.

TABLE VII

PERFORMANCE COMPARISONS ON HRSC2016

Methods	Backbone	mAP(07)	mAP(12)
RoI Transformer [16]	R-101	86.20	-
CenterMap-Net [59]	R-50	-	92.80
RetinaNet-O [13]	R-101	89.18	95.21
FAA [32]	R-50	89.80	95.90
ABFL [71]	R-50	89.98	-
TIOE-Det [67]	R-101	90.16	-
S^2ANet [19]	R-101	90.17	95.01
TIR-Net [72]	R-50	90.20	-
HA-RDet [45]	-	90.20	95.32
AOPG [54]	R-101	90.34	96.22
DSSNet [22]	ConvNeXt-T	90.38	97.26
O-RCNN [10]	R-50	90.40	96.50
DynaPro [73]	HiViT-B	90.40	97.72
ReDet [29]	ReR-50	90.46	-
QPDet [74]	R-50	90.47	96.60
UFDet [65]	R-50	90.47	97.50
Ours	R-50	**90.50**	**97.90**

4) Computational Efficiency Evaluation: Table VIII presents a comparison of the accuracy and efficiency of our method with those of other methods. Compared to the baseline model, the complete model adds only 3.3 M parameters and 2 GFLOPs, resulting in a minimal decrease in FPS but a 3.92% improvement in mAP. Although LSKNet-S [47] and O^2-DFINE [28] achieve competitive mAPs with fewer parameters, mAP alone does not fully reflect the

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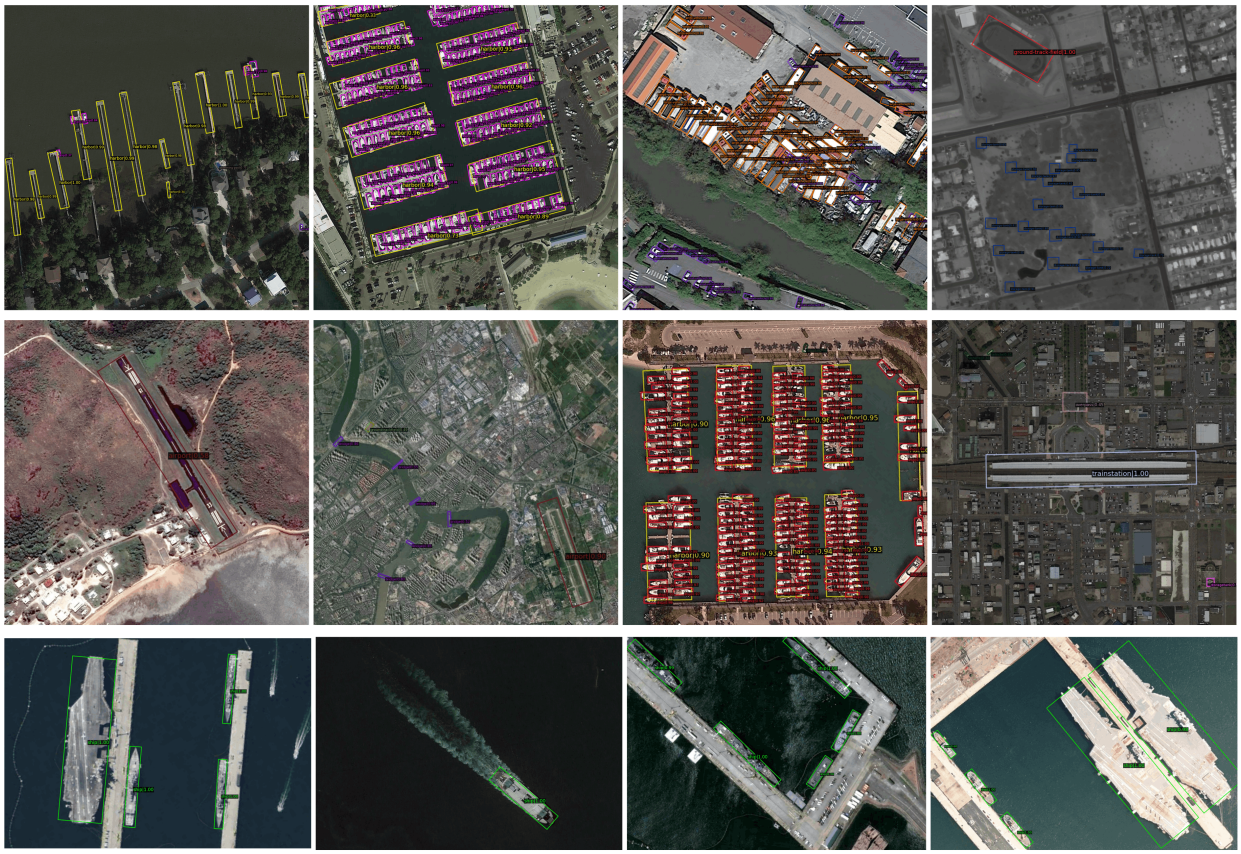


Fig. 5. Qualitative visualization of the proposed method.

TABLE VIII
COMPARISON OF COMPUTATIONAL EFFICIENCY WITH OTHER DETECTORS

Methods	#P	FLOPs	FPS	mAP(%)
Oriented RCNN [10]	41.1M	211G	38.8	73.95
ARC [30]	74.4M	217G	-	77.35
LSKNet-S [47]	**31.0M**	**161G**	-	77.49
O^2-DFINE [28]	**31.0M**	229G	132.0	77.73
DynaPro [73]	147.7M	583G	-	77.85
Ours	44.4M	213G	38.0	**77.87**

advantages of our method. As shown in Table V, our method achieves competitive results on categories such as harbor (HA), ship (SH), and soccer ball field (SBF). From a practical deployment perspective, LSKNet-S and O^2-DFINE are indeed more parameter-efficient for general-purpose detection, but our method addresses the challenges of modeling continuous orientation and handling extreme ARs. Furthermore, given the moderate increase in parameters relative to the baseline, the improvement in detection quality for difficult categories is reasonable. In summary, the method proposed in this article significantly improves detection accuracy with only a slight increase in computational cost, achieving a good balance between accuracy and efficiency.

D. Visualization Result

Fig. 5 shows the qualitative detection results of the proposed INFF-Net on three benchmark datasets. The first row presents the detection results for the DOTA-v1.0 dataset. Our method achieves accurate detection even under complex conditions such as poor imaging quality, low contrast, and dense object clusters while also demonstrating good detection performance for objects with extreme ARs and arbitrary orientations in port areas. The second row shows the detection results for the DIOR-R dataset. Despite the presence of complex backgrounds, objects with extreme ARs (such as airports and dams), and densely overlapping ships near the harbor, our method still achieves good detection performance. The third row shows the detection results on the HRSC2016 dataset; our method is able to accurately locate slender, objects with high ARs. These results collectively validate the effectiveness and strong generalization capability of the proposed method.

Fig. 6 shows the visualization results comparing our method with the baseline. As can be seen, the baseline method suffers from significant missed detections, and some detection boxes are misaligned with the actual object contours. In contrast, the detection boxes generated by our method closely follow the object contours, with virtually no missed or false detections for densely packed objects. The significant improvement in localization accuracy demonstrates our method's robust feature representation capabilities for objects with varying orientations while also validating its distinct advantages in detecting objects with extreme ARs and in complex backgrounds.

As shown in Fig. 7, the first two images present class confusion: the bridge is misidentified as dam and the overpass as bridge. The last two exhibit partial missed detection in

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Fig. 6. Visualized results compared to the baseline. (a) Image. (b) GT. (c) Baseline. (d) Ours.



Fig. 7. Failure case visualization. (a) Class confusion. (b) Extremely dense small objects.

dense scenes, where some storage tanks and harbor ships are undetected due to occlusion and overlapping features of neighboring objects. For objects with ambiguous semantic boundaries, the model may still exhibit class confusion due to their highly similar geometric structures and contextual environments. This reflects that the implicit feature field may lack the ability to model fine-grained semantic differences between objects and lacks contextual semantic inference. In harbor scenes with densely packed ships and partial occlusions, a small number of missed detections still occur. This may be attributed to severe feature overlap between objects in extremely dense scenes, where the importance-guided sparse sampling mechanism faces accuracy bottlenecks when distinguishing the boundaries of adjacent objects.

V. Conclusion

This article addresses three major bottlenecks in the detection of rotated objects in remote sensing—the lack of rotation-aware representations, redundant background noise, and inadequate modeling of objects with extreme ARs—by proposing a rotation-aware remote sensing object detection

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method based on INFF-Net. This method constructs an IRFF to achieve a rotation-aware continuous feature representation regularized by soft rotation-equivariance constraints. It employs an ISSM module to enhance the signal-to-noise ratio of RoI features and introduces an ASFE module to optimize the detection performance for objects with extreme ARs. Experimental results on the DOTA-v1.0, DIOR-R, and HRSC2016 datasets demonstrate that our method achieves improved detection accuracy compared to most mainstream approaches, demonstrating its effectiveness. Ablation studies further validate the synergistic effects of the three modules. Future work will explore the incorporation of fine-grained semantic modeling and context-aware reasoning mechanisms to improve the accuracy of object boundary detection in scenes with dense occlusions.

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